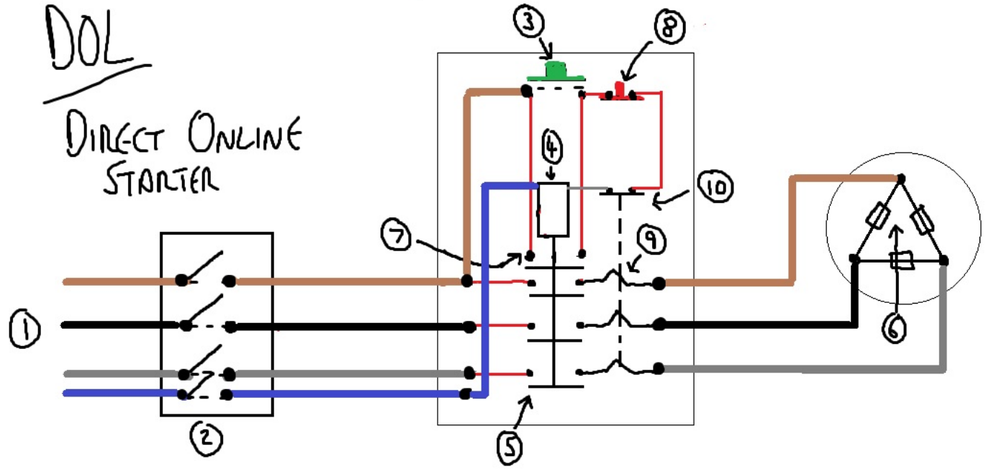


What happens when 2 and 3 are operated?

The isolator (2) when closed allows current to flow into the contactor. The start button (3) is normally open, when this is closed the current flows into the coil (4) energising it. The coil magnetises the main contacts and pulls them closed and holds them in place, enabling the motor to connect to the supply and become energised.

What happens when 8 is operated?

8 is a normally closed stop button. It works on the opposite principle of the start button. The start button 'makes' the circuit, the stop button 'breaks' the circuit. Once operated, the stop button disconnects the circuit which de-energises the coil, releasing the contacts and isolating the motor from the supply.

What is the purpose of 9 and 10?

9 and 10 work together to disconnect the circuit in the same way as the stop button. The thermal overload trips are simply bi-metallic strips which will bend when too much heat is passed through. The bend in the strips causes the overload release contact to break the circuit, which de-energises the coil, releasing the contacts and isolating the motor from the supply.

What is the purpose of 4?

Inside the contactor coil is a solenoid. Current energises the coil which magnetises the solenoid, which in turn pulls the latching contacts closed. If the current is interrupted (from either 8 or 9/10) the could de-energises, the magnetic effect disappears and the latching contacts release.

Component list:

1) 3-phase and neutral supply
2) 4 pole isolator
3) Normally open (N/O) start button. This is 'sprung' meaning it'll return to N/O once released.
4) 230V contactor coil (only connected to one phase and neutral meaning only 230V is present.
5) Main contacts. These join the supply to the load once 'pulled' in and held.
6) Delta connected load.
7) Latching (maintaining) contacts. These 'pull' and 'hold' the main contacts in place, these are pulled into place by the magnetic field created by the energising of (4).
8) Normally closed (N/C) stop button
9) Thermal overload trip
10) Overload release contact

Where are these starters used and what are their limitations?

1) Generally used for the starting and stopping of small machinery up to 5kW.
2) Compressors, pumps and fans are typically controlled by DOL starters.
3) Due to the nature of the starter, a high in-rush of current can be tolerated, unlike a traditional switch.
4) DOL starters have thermal overloads, which can mean 'overuse' or regular stopping and starting can cause these to operate.